

GHK-Cu

Copper Peptide | 50mg & 100mg x 10 Vials

Skin Repair | Collagen Synthesis | Anti-Inflammatory | SubQ / Topical-Adjacent

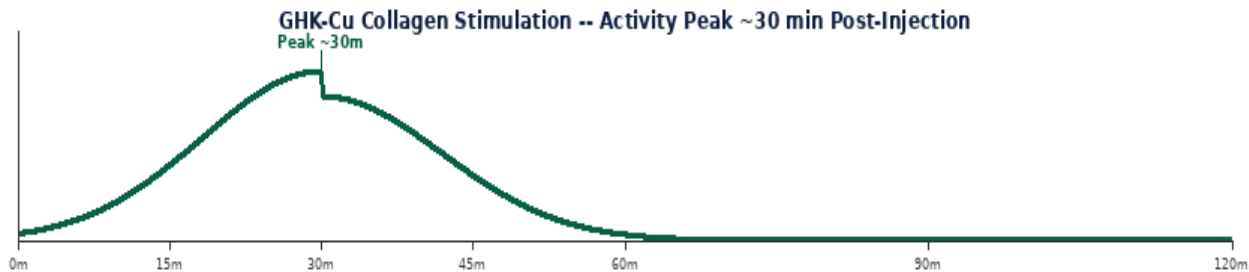
Purity: 99.8% | Endotoxin-Free | Research Use Only

GHK-Cu (Glycyl-L-histidyl-L-lysine copper complex) is an endogenous tripeptide naturally present in human plasma, urine, and saliva. It binds copper (Cu2+) and exerts potent regenerative, anti-inflammatory, and collagen-stimulating effects. Endogenous levels decline sharply with age. Injectable GHK-Cu achieves systemic concentrations not possible with topical formulations.



Parameter	Specification
Class	Copper-binding tripeptide -- regenerative, anti-inflammatory, collagen-stimulating
Vial Sizes	50mg x 10 100mg x 10 Purity 99.8%
Reconstitution	50mg: 1mL BAC Water->50mg/mL 100mg: 2mL BAC Water->50mg/mL
Standard Dose	500-5,000 mcg (0.5-5mg) per injection SubQ
Frequency	1-2x daily SubQ no fasting required
Key Benefits	Collagen synthesis, wound healing, skin repair, anti-inflammatory, neuroprotection
Cycle & Storage	4-12 weeks on, 2-4 weeks off 2-8 deg C use within 28 days

MECHANISM OF ACTION & RESEARCH BENEFITS



Pathway	Mechanism	Benefit
Collagen & Wound Healing	Activates TGF-beta and downstream collagen type I/III synthesis; promotes fibroblast migration and proliferation; accelerates wound closure; stimulates angiogenesis via VEGF upregulation	Wound healing, scar remodelling, skin repair, post-procedure recovery
Anti-Inflammatory & Antioxidant	Downregulates NF-kB inflammatory cascade; reduces TNF-alpha, IL-6; copper-superoxide dismutase (Cu/Zn-SOD) activation; neutralises reactive oxygen species; reduces oxidative stress	Systemic anti-inflammatory; antioxidant; tissue protection
Neuroprotection & Gene Modulation	GHK-Cu modulates expression of >4,000 human genes; promotes nerve regeneration (NGF-like activity); antifibrotic effects in lung and liver; resets gene expression toward a younger phenotype	Anti-aging, neurological research, organ fibrosis, gene expression modulation

Benefit	Context
Collagen Synthesis	GHK-Cu is one of the most potent natural stimulators of collagen I and III production, with clinical applications in wound healing and anti-aging.
Gene Regulation	Research shows GHK-Cu modulates expression of over 4,000 human genes, resetting many toward a younger expression profile.
Anti-Aging & Skin	Endogenous GHK-Cu declines from ~200ng/mL at 20 years to ~80ng/mL at 60. Injectable supplementation restores youthful levels systemically.

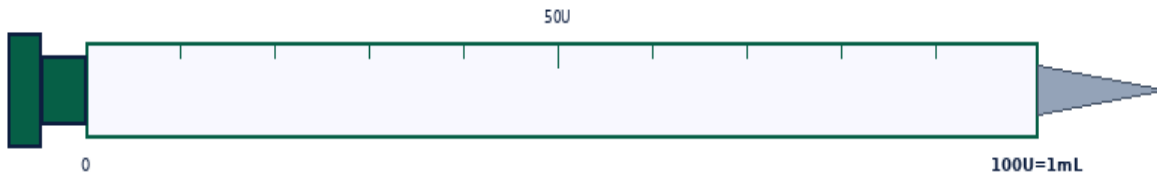
RECONSTITUTION & DOSE CALCULATION

All vials use 1 mL BAC Water. Room temperature. Swirl gently -- never shake.



1 mL BAC Water per vial. Roll gently -- NEVER shake.

Step	Action	Detail
1	Gather	Vial, BAC Water, insulin syringe (29-31G), swabs, sharps.
2	Clean stoppers	Wipe both stoppers. Air-dry 10-15 sec.
3	Draw 1mL BAC	Exactly 1 mL. Expel air bubbles.
4	Inject into vial	Aim at GLASS WALL. Slowly. Never onto powder.
5	Swirl & label	Roll 20-30 sec. Clear within 60 sec. Never shake. Label date. 2-8 deg C. Use within 28 days.



1mL Insulin Syringe | 29-31G | SubQ

Formula: Vol(mL)=Dose(mcg)/Conc(mcg/mL) | **Units:** Vol x 100

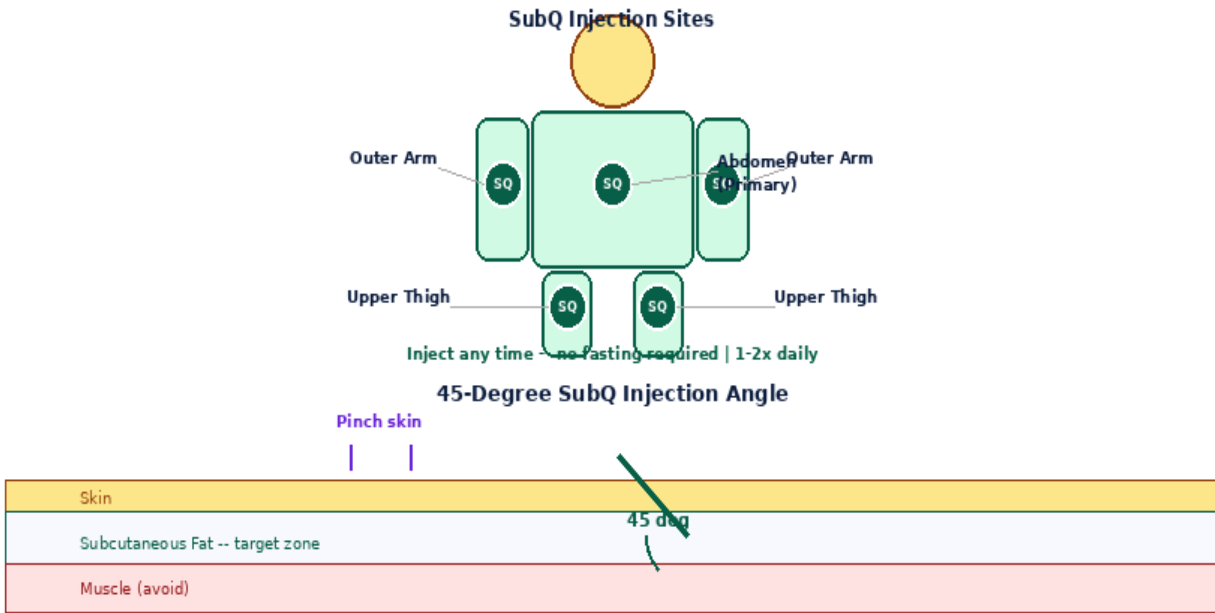
50mg x 10 -- 1mL BAC -> 50.0mg/mL (50,000mcg/mL)

Dose(mcg)	500mcg	1000mcg	2000mcg	3000mcg	5000mcg
Vol(mL)	0.0100	0.0200	0.0400	0.0600	0.1000
Units	1.0U	2.0U	4.0U	6.0U	10.0U
Doses/Vial	100	50	25	16	10

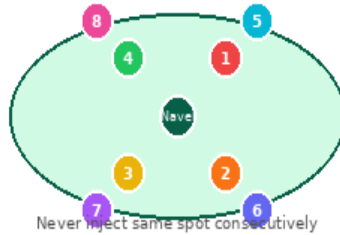
100mg x 10 -- 2mL BAC -> 50.0mg/mL (50,000mcg/mL)

Dose(mcg)	500mcg	1000mcg	2000mcg	3000mcg	5000mcg
Vol(mL)	0.0100	0.0200	0.0400	0.0600	0.1000
Units	1.0U	2.0U	4.0U	6.0U	10.0U
Doses/Vial	100	50	25	16	10

INJECTION TECHNIQUE, PROTOCOLS & GOLDEN RULES



8-Point Rotation Map -- Rotate Every Injection



Step	Action	Detail
1	Wash & prepare	Wash hands. 29-31G insulin syringe. No fasting required.
2	Draw dose	Draw exact volume per dose table. Expel air bubbles.
3	Inject SubQ	Abdomen preferred. Pinch 2-3cm. 45 deg. Inject slowly.
4	Rotate & dispose	8-point rotation. Press 5 sec. Sharps container.

Protocol	Dose	Frequency	Duration	Notes
Skin/ Anti-Aging	1,000-2,000 mcg	1-2x daily	8-12 wk 2-4 off	No fasting. Any time.
Wound Healing	2,000-5,000 mcg	1-2x daily	4-8 wk	Higher dose for acute repair protocols.

Side Effect	Likelihood	Management
Injection site reaction	Common	Mild redness at site. Rotate sites.
Temporary skin darkening	Occasional	Copper-mediated melanin; resolves. Not harmful.
Nausea at high doses	Rare	Reduce dose. Inject with food.

Contraindications: Cancer history, pregnancy, under 18.

1. 50mg vial: 1mL BAC Water yields 50mg/mL. 100mg vial: 2mL yields 50mg/mL.
2. No fasting required -- GHK-Cu is not dependent on GH/insulin axis.
3. Start at 500-1,000 mcg and escalate based on response and tolerance.
4. Inject SubQ into abdomen or other SubQ sites. Rotate 8-point map.
5. Can be injected 1-2x daily. Morning or bedtime -- no timing restriction.
6. Store at 2-8 deg C. Use within 28 days of reconstitution. Never freeze.
7. Cycle 4-12 weeks. Short cycles preferred for wound/skin protocols.
8. Safe profile at standard doses. Contraindicated in Wilson disease (copper overload).
9. Not for use during pregnancy, lactation, or in individuals under 18.

DISCLAIMER:

research use only. Not for human therapeutic use. Consult a qualified professional.

assumes no liability.

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